

HRMS — TWO-PHASE ANTIGRAVITY BUILD SPECIFICATION

Condensed execution specification • September 2026

Purpose: Give this PDF directly to Antigravity as the implementation contract. The original multi-stage roadmap is compressed into exactly **2 execution phases**. All production-scope HRMS capabilities remain in scope; only the sequencing is simplified.

Build rule: Do not create additional development phases. Do not omit a listed feature. Build Phase 1 completely, then Phase 2 completely. Configurable policies, statutory rules, entitlements and customer settings must be data-driven rather than hard-coded.

1. PRODUCT CONTRACT

Build an India-first, internal-first, multi-tenant SaaS HRMS for startups, small businesses and SMBs, initially supporting approximately 100 employees per tenant. The product covers: recruit → hire → onboard → manage → attend → leave → pay → develop → reimburse → manage assets → exit.

Contract	Required
Architecture	Modular monolith with strict tenant isolation; clear module/service/repository/event/API boundaries.
Frontend	Next.js + TypeScript; Tailwind CSS + reusable component library; responsive web first.
Backend	NestJS + TypeScript; DTO validation, guards, services/use-cases and repositories.
Database	PostgreSQL + Prisma; relational business data; JSONB only where appropriate.
Jobs / Cache	Redis + BullMQ; payroll, imports, notifications and AI jobs must be observable, retryable and idempotent.
Storage	Private S3-compatible object storage; signed URLs; metadata in PostgreSQL.
Auth	Email/mobile + password V1; revocable sessions/tokens; MFA-ready; SSO-ready.
Communications	In-app + email V1; WhatsApp provider abstraction; push-ready for later mobile.
PDF / Reports	Server-side PDF generation; CSV/XLSX/PDF exports according to permissions.
Observability / CI	OpenTelemetry + central logs; GitHub Actions CI/CD.
Deployment	Local → development/shared test → staging/UAT → production; AWS recommended.

2. TWO-PHASE DELIVERY MODEL

Phase	Scope	Exit gate
PHASE 1 — Foundation + Security	Platform foundation + organization/employee master + roles + employee portal + attendance + leave management + payroll + statutory compliance	Initial attendance and leave management; payroll and statutory compliance
PHASE 2 — Full Lifecycle + Reimbursement + Production	Recruitment + hiring + onboarding + documents/e-sign + performance + expenses + assets + fulfillment + walk reports end + internal helpdesk + security	Full lifecycle HRMS capabilities; internal helpdesk and security

The two phases absorb the original architecture, implementation, pilot and hardening stages without removing their requirements.

PHASE 1 — FOUNDATION + CORE HR + PAYROLL

Goal: Establish the secure multi-tenant platform and deliver the complete core HR operating system, including attendance, leave, approvals and payroll/statutory compliance.

1A. Platform foundation

- Repository/branching, TypeScript strict mode, lint/type-check/test/build pipeline and migration workflow.
- Tenant lifecycle: signup → tenant → admin → payment method → 30-day trial → setup wizard → active → grace/restricted → suspended → archive/retain.
- Tenant isolation: tenant_id on tenant data; tenant context from authentication; automatic tenant filtering; tenant-scoped jobs/files/exports/events/webhooks.
- Authentication: email/mobile + password; invited/active/locked/suspended/deactivated/archived states; password reset; rate-limited login; audit login events; MFA-ready.
- RBAC: Platform Super Admin, Platform Support Admin, Company Admin, HR Admin, HR Executive, Finance, Manager, Employee and Custom Role.
- Permission keys and data scopes: own records, direct reports, department, branch/location, legal entity, entire tenant.
- Separate audited sensitive permissions for salary/payroll, bank, identity, tax, face verification, medical/sensitive data, exports, audit logs and API credentials.
- Generic approval workflow engine with backup approvers; modules must reuse it.
- Audit logs: tenant, actor, role, action, entity, timestamp, source, request ID and before/after values for sensitive mutations.
- Private object storage, signed URLs, document metadata, validation/scanning architecture, versions, checksums and retention.

1B. Organization + employee master

- Hierarchy: legal entities → branches → locations → departments/teams; designations and grades.
- Employee lifecycle: draft → invited/onboarding → active → probation → confirmed → transferred → notice → exited → archived.
- Employee master: identity, personal, contact, employment, organization, payroll, security and audit fields; contacts, addresses, emergency contacts, bank, tax, documents and reporting manager.
- Employee ID pattern engine: prefix, year, branch/location, sequence and zero-padding; tenant-unique IDs.
- Custom employee fields: text, number, date, dropdown, multi-select, boolean, currency, long text, file and entity reference with required/visibility/validation/export controls.
- CSV/Excel import with validation, actionable errors and duplicate handling.

1C. Attendance + time management

- Web check-in/out, GPS, geofencing, QR attendance, configurable selfie/face verification and biometric-device adapter architecture.
- Attendance state machine: not started → check-in → worked → optional break → check-out/completed; exceptions for missing punch, regularization, WFH, on-duty, holiday, week-off and absent.
- Policies: work days, weekly offs, shift start/end, grace, late/early thresholds, half-day, overtime, breaks, auto-punch, missing punch, cross-midnight, week-off/holiday work, WFH/on-duty, regularization cutoff, geofence radius and GPS accuracy.
- Immutable raw events; daily processor derives attendance from raw events, shifts, holidays, leave, WFH/on-duty and overrides.
- Regularization request with proposed times/reason/attachment; manager → backup approver; approval updates processed attendance without deleting raw events.

1D. Leave management

- Casual, Sick, Earned/Privilege, Maternity, Paternity, Comp Off, Loss of Pay, WFH and custom leave types.
- Policy engine: accrual, opening balance, carry-forward/cap, expiry, encashment, half-day, negative balance, probation, weekend/holiday behavior, sandwich rule, documents, consecutive-day limits, notice period, blackout dates and eligibility.
- Leave request → validation → workflow → manager/backup approver → decision → ledger → attendance impact → notification.
- Ledger transactions: OPENING, ACCRUAL, CONSUMPTION, REVERSAL, ADJUSTMENT, EXPIRY, ENCASHMENT; displayed balance is derived from the ledger.

1E. Payroll + statutory compliance

- Salary structures with flat, percentage and conditional formulas; effective dating; assignments; revision history; arrears; CTC versus earning/deduction views.

- Earnings: Basic, HRA, special allowance, conveyance, variable pay, incentives, bonus, overtime, arrears, reimbursements.
- Deductions: PF, ESI, professional tax, TDS, LOP, recoveries and custom deductions; employer PF/ESI/gratuity and other employer costs.
- Pipeline: select period → freeze eligibility → salary → attendance/LOP/leave → one-time items → statutory rules → TDS → adjustments → validation → review → approval → lock → payslips → notifications → publish.
- Versioned statutory rules: effective dates, jurisdiction, eligibility, wage/base, thresholds, rate/formula, rounding, limits, exemptions, source, version, active flag and publish approval.
- Old and new income-tax regimes; fiscal-year versioning; declarations/deductions/exemptions; annual projection and monthly TDS. Separate EPF, ESI, professional-tax and TDS rule families.
- Authorized manual overrides require employee/run/component, old/new amount, reason, actor and timestamp; original calculation remains traceable.
- Payroll state: DRAFT → CALCULATED → REVIEW → APPROVAL_PENDING → APPROVED → LOCKED → PUBLISHED; controlled reopen/reject path.
- Payslips include company/legal entity, employee, period, payable days, earnings, deductions, employer contributions if enabled, gross, net, tax, YTD, number and generation timestamp.
- Payroll must be deterministic, reproducible, transactional, auditable and idempotent; locked runs cannot be edited by normal users.

1F. Employee / Manager / HR experience

- Employee dashboard: attendance today, check-in/out, leave balance, holidays, pending requests, latest payslip, notifications and announcements.
- Employee modules: profile, attendance, leave, payslips, documents, expenses, assets, performance, requests, notifications and directory.
- Manager dashboard: team attendance, leave approvals, regularizations, WFH/on-duty, birthdays/anniversaries, appraisals, team list and quick actions.
- HR dashboard: headcount, present/on leave/WFH/absent, joiners, exits, payroll status, onboarding, document verification, recruitment and performance status.
- Directory privacy configurable; personal phone, bank, salary, identity and sensitive data never exposed.

PHASE 1 ACCEPTANCE GATE

- AI Automation Labs tenant created and isolated; real employees imported successfully.
- Roles, permissions and data scopes enforce access correctly.
- Attendance runs through at least one complete payroll cycle, including policies and regularization.
- Leave policies, ledger, approvals and backup approver routing work.
- Payroll calculates, reconciles, approves, locks and generates payslips without duplicate results.
- Automated unit, integration/API, UI, migration, permission, tenant-isolation and payroll tests pass.

PHASE 2 — FULL LIFECYCLE + AI + SAAS + PRODUCTION

Goal: Complete every remaining production-scope capability and harden the system into an internal pilot and controlled multi-tenant SaaS product.

2A. Recruitment / ATS

- Manpower requisition → approval → job requisition → publish → application → screening → shortlist → interview → feedback → selection → offer → acceptance → joining → employee creation.
- Public careers page; job descriptions; candidate profiles; resume upload/parsing; pipeline; interview scheduling/panels; feedback forms; offers; candidate communication.
- AI job matching/scoring, duplicate detection, interview-question generation and feedback summarization.
- Candidate data: identity/contact, location, education, experience, skills, resume, current employer/compensation, notice period, source, consent/privacy, interview/application/offer history.

2B. Onboarding

- Reusable templates by employee type/department/location/job family; task owner, due date, dependency, required flag, attachment requirement and evidence.
- Offer accepted → onboarding instance → profile → identity/tax/bank documents → appointment/policy signing → IT setup → asset assignment → manager induction → HR verification → activation.

2C. Documents + internal e-sign

- Offer, appointment, experience, relieving, increment, promotion, warning, confirmation, internship, payslip and custom HR documents.
- Versioned templates with employee/company merge fields; generated documents retain template version.
- Internal signing with signer/document/deadline/status, sent/opened/signed/declined/expired audit trail, immutable signed version, optional OTP/email verification and signature placement.
- External e-sign provider abstraction remains ready.

2D. Performance management

- Monthly, quarterly, half-yearly, annual and custom cycles; simultaneous cycles by team/employee.
- Cycle setup → goals → self appraisal → manager appraisal → optional 360 → final rating → promotion/increment recommendation → approval → closure.
- KRAs, KPIs, OKRs, goal weights, ratings, PIP, recommendations, calibration/finalization and historical archive.

2E. Expenses

- Claims with category, amount, date, description, receipts, policy validation, project/client/cost center, mileage/travel and future tax/GST fields.
- Manager approval → Finance approval → payment status; bank transfer, cash, payroll reimbursement or configurable payment method.
- Employee/team views, approval queue, finance queue and reimbursement status.

2F. Assets

- Asset master/category, serial/asset number, purchase, warranty, location, assignment, dates, condition, repairs, loss/damage and acknowledgement document.
- Asset list, assign, return and history; unreturned assets feed reports and exit clearance.

2G. Exit + Full & Final Settlement

- Resignation/HR initiation → approval → LWD → clearance → asset return → HR/Finance/IT clearance → attendance/leave freeze → F&F; → payroll approval → final documents → deactivation → archive.
- F&F;: salary, LOP, leave encashment, notice recovery/payout, bonus/incentives, reimbursements, deductions/recoveries, employer liabilities where applicable, final amount, F&F; statement, relieving and experience letters.

2H. Notifications + communications

- In-app/email for invitations, onboarding, workflows, leave, payroll and tasks; WhatsApp provider abstraction; future push support.

- Template management, merge variables, per-event controls, role-based recipients, delivery state, retries, provider failure handling, opt-out where appropriate and audit history.

2I. AI service layer

- Controlled tenant-scoped AI gateway; never bypass RBAC.
- Employee assistant: leave balance, salary explanation, payslip location, pending requests and leave application preparation; mutations require explicit confirmation.
- HR copilot: natural-language employee search, attendance anomalies, leave patterns, probation reminders, attrition/headcount summaries, document generation and policy Q&A; from approved company documents.
- Recruitment AI: resume parsing, JD extraction, matching/scoring, duplicate detection, interview questions and feedback summaries.
- Payroll assistant: salary/deduction/LOP explanations, component breakdown and unusual variance flags.
- AI safety: no unauthorized salary data; no unconfirmed salary/bank/tax/statutory mutation; AI cannot approve its own workflow; minimize sensitive external-AI data; log AI requests/confirmations; tenant AI enable/disable.

PHASE 2 — PLATFORM, ANALYTICS AND PRODUCTION

2J. API + Webhooks

- REST /api/v1; bearer/API-key auth; JSON; pagination; structured errors with request_id; ISO 8601 dates; documented money convention.
- Core APIs for auth, employees/import, attendance/check-in/out/regularization, leave/approval, payroll/calculation/approval, payslips, jobs, candidates, expenses, performance cycles, exit cases and webhooks.
- Webhook events: employee created/updated/exited; attendance check-in/out/regularization; leave created/approved/rejected; payroll processed; payslip generated; candidate created/hired; expense approved; document signed; workflow task created.
- HTTPS + HMAC tenant secret + timestamp/replay protection + delivery ID + exponential retry + dead-letter + logs + manual replay.

2K. Reports + analytics

- Employee: master, headcount, joiners, exits, department/location.
- Attendance: daily/monthly, late, early departure, absenteeism, overtime, regularization.
- Leave: balances, utilization, department analysis, trends, pending requests.
- Payroll: summary, salary register, deductions, PF/ESI/TDS/PT extracts, payslip register, F&F.;
- Recruitment: open roles, pipeline, source effectiveness, time-to-hire, interview outcomes.
- Performance: cycle status, rating distribution, goal completion, PIP, promotion/increment recommendations.
- Expenses: register, approvals, reimbursement status, category spend.
- Assets: register, employee assets, unreturned assets, warranty/expiry.
- Management: headcount trend, attrition, payroll cost, attendance trend, leave trend.
- Reports support date range, organization/department/branch/location filters where meaningful and permission-controlled CSV/XLSX/PDF export with audit logging.

2L. SaaS Super Admin + support

- Tenant onboarding, suspension/reactivation, trial/subscription state, usage, employee counts, entitlements, storage/API/AI/WhatsApp usage, support tickets, audit logs, system health, configuration flags, statutory-rule publishing, template library and audited impersonation.
- Super-admin dashboard: active/trial/expiring/suspended tenants, employee count, API errors, job failures, notification failures, storage and open support tickets.
- Support impersonation must show an unmistakable banner and audit support user, tenant, target user, reason, start/end time; secrets remain inaccessible.

2M. Subscription + trial

- Configurable plans/entitlements; prices never hard-coded.
- Plan controls: employee limit, module access, storage, API/webhooks, WhatsApp allowance, AI allowance, admin count, locations, support tier and feature flags.
- Signup → payment-method token → 30-day trial → reminders → expiry → grace/restricted → suspension if unpaid → reactivation.

2N. Customization + branding

- Configurable departments, designations, leave types/policies, attendance rules, working hours, shifts, holidays, salary components, workflows, employee custom fields, document templates, notifications, employee ID formats, onboarding checklists, performance cycles and expense categories.
- AI Automation Labs branding remains visible/compulsory; tenant logo/primary color can appear on portal, payslips, letters, reports and PDFs while platform attribution remains.

2O. Security, privacy, retention + files

- TLS, encryption at rest where supported, secure hashing, MFA-ready controls, login alerts, session/device revocation, rate limiting, API auth, RBAC, tenant isolation, backups, disaster recovery, retention, soft delete, secret management and dependency scanning.
- CSRF/XSS/SQL-injection protection; upload validation and malware-scanning architecture.
- Aadhaar/PAN/identity, bank, salary, tax, face-verification and medical/sensitive fields are masked/restricted and audited on access/exports.
- Configurable retention by data category; deletion respects legal retention and audit requirements.

- Document lifecycle: upload → validate → scan → store → verify/sign → active → replace/expire → archive/delete.

2P. Reliability + DevOps

- Fast measured/monitored authenticated pages and APIs; initial scale ~100 employees/tenant with growth-ready architecture.
- Health checks, monitoring, restart strategy; transactional payroll/leave/financial writes.
- Background jobs use idempotency keys, persisted status, retry transient failures, dead-letter permanent failures and tenant/actor/source context.
- CI/CD: git push → lint → type check → unit tests → integration tests → build → security → migration validation → staging → smoke tests → approval → production.
- Production schema changes only through migrations; reversible or documented recovery; staging backup-data validation; chunk large migrations; payroll-impacting schema changes require regression tests.

2Q. Complete QA / Definition of Done

- Test levels: unit, service/integration, API, migration, UI/component, end-to-end, permission, tenant isolation, payroll, import/export, notification, security, performance and regression.
- Critical tests: cross-tenant denial; locked-login block/audit; duplicate import rejection; geofence denial; raw punch preservation; leave-policy enforcement; backup approver; idempotent payroll; audited overrides; locked payroll immutability; correct document template version; immutable signed document; invalid API-key denial; webhook retry/dead-letter; AI salary-data denial; explicit AI leave confirmation; expense payment state; exit deactivation + F&F; start.
- Module Definition of Done: functional requirements, permission matrix, tenant isolation, audit events, unit/integration tests, UI loading/error/empty states, validation, import/export behavior, idempotent jobs where applicable, monitoring/logging, migration test, security review, help text where needed and acceptance criteria.

3. MASTER FEATURE COVERAGE — NOTHING OMITTED

This checklist is the condensed implementation inventory. Every production-scope item belongs to Phase 1 or Phase 2.

Area	Included capabilities
Identity & Access	Login, reset, account states, MFA-ready, sessions/devices, RBAC, custom roles, data scopes, sensitive permissions, audit.
Multi-Tenant SaaS	Tenant lifecycle, isolation, scoped jobs/files/events/webhooks, support impersonation, trials, plans, entitlements, feature flags.
Organization	Legal entities, branches, locations, departments, teams, designations, grades, holidays, employee ID configuration.
Employee	Master, contacts, addresses, emergency contacts, bank, tax, documents, custom fields, reporting, employment history, lifecycle.
Attendance	Web, GPS, geofence, QR, selfie/face configurable, biometric-ready, shifts, policies, processing, regularization.
Leave	Standard/custom types, accrual, balances, carry-forward, expiry, encashment, negative balance, eligibility, blackout/sandwich rules, approvals, ledger.
Payroll	Salary structures, formulas, revisions, arrears, earnings, deductions, employer costs, statutory rules, PF/ESI/PT/TDS, regimes, overrides, approvals, locking.
Recruitment	Requisitions, jobs, careers, candidates, resumes, parsing, pipeline, interviews, feedback, offers, communications, hiring conversion, AI matching/scoring.
Onboarding	Templates, instances, tasks, dependencies, profile/doc collection, signing, IT setup, assets, induction, verification, activation.
Documents / E-Sign	Templates, versioning, generated docs, verification, internal signing, audit, immutable signed versions, provider abstraction.
Performance	Cycles, KRAs/KPIs/OKRs, goals/weights, self/manager/360, ratings, PIP, calibration, promotion/increment, archive.
Expenses	Claims, receipts, policy validation, project/client/cost center, travel/mileage, approvals, finance, payment/reimbursement.
Assets	Master, categories, serials, purchase, warranty, location, assignment/return, condition, repairs, loss/damage, acknowledgement.
Exit / F&F	Resignation, approval, LWD, clearance, asset return, freeze, F&F, payroll approval, final docs, deactivation, archive.
Self-Service	Dashboard, profile, attendance, leave, payslips, docs, expenses, assets, performance, requests, notifications, directory.
Dashboards	Employee, Manager, HR and Super Admin dashboards.
Notifications	In-app/email, WhatsApp abstraction, templates, variables, event controls, recipients, delivery/retry/failure/audit.
AI	Employee assistant, HR copilot, recruitment AI, payroll explanation/variance, policy Q&A, confirmations and safety controls.
API / Webhooks	REST API, API keys, core endpoints, events, HMAC, replay protection, retries, dead-letter, logs/replay.
Reports	Employee, attendance, leave, payroll/statutory, recruitment, performance, expenses, assets, management; CSV/XLSX/PDF.
Security / Storage	Encryption, hashing, MFA-ready, rate limiting, headers, CSRF/XSS/SQLi protection, upload validation/scanning, private storage, signed URLs, retention.
DevOps	Local/dev/staging/prod, CI/CD, migrations, background jobs, observability, backups, recovery, health checks, performance/security hardening.

4. NON-NEGOTIABLE DATA / ARCHITECTURE RULES

- Core entities cover tenants, legal entities, branches, locations, departments, teams, designations, grades, users, roles, permissions, employees, attendance, leave, payroll, ATS, onboarding, documents/signatures, performance, expenses, assets, exits/F&F;,, notifications, workflows, API keys, webhooks, audit logs, support tickets, subscriptions and feature flags.
- Common fields: UUID id, tenant_id where applicable, created/updated timestamps and actors, status/state, soft-delete where appropriate, optimistic-lock version where needed.
- Business-critical searchable fields stay relational; JSONB is for configuration snapshots, custom fields and provider payloads.
- Services/use-cases contain business logic; repositories/services contain DB access; controllers stay thin.
- Every mutation validates authorization; every tenant query carries tenant context; multi-table financial/HR changes use transactions.
- No secrets or unnecessary sensitive data in logs; use request/job IDs and tenant IDs.
- No hard-coded statutory rates or customer policies; use configurable/versioned rules with effective dates and source references.

5. INTERNAL-FIRST RELEASE GATE

- AI Automation Labs tenant created; real employees imported.
- Attendance used for at least one full payroll cycle.
- Leave workflows used internally.
- Payroll manually reconciled before release.
- Documents generated and signed.
- Expenses, assets and exit/F&F; tested.
- Security and tenant-isolation tests pass.

- Management approves pilot readiness before controlled SaaS beta.

6. FUTURE-READY, NOT PART OF THE TWO-PHASE CORE DELIVERY

Keep architecture ready for native Android/iOS, biometric device integrations, Google/Microsoft SSO, external e-sign providers, accounting integrations including Tally/Zoho Books/QuickBooks, OpenSearch, enterprise SSO, custom domains/expanded white-labeling, multi-country payroll, multi-currency, multi-language, advanced benefits, engagement surveys, learning management, travel management, project timesheets, shift marketplace/swaps and advanced workforce analytics.

7. ANTIGRAVITY EXECUTION COMMAND

BUILD ONLY TWO PHASES. Implement PHASE 1 completely, validate its gate, then implement PHASE 2 completely. Internal dependency order is allowed, but do not turn dependencies into additional user-facing phases. Do not omit features from the coverage matrix. Do not wait for another specification. Treat this PDF as the master execution contract.

Source: AI Automation Labs LLP — HRMS Master Product Requirements & Developer Specification, Version 1.0, September 2026. This PDF compresses the original roadmap into two execution phases while retaining the production-scope feature set.